

ALEXANDER BLOOM

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EDUCATION

B.S. Electrical and Computer Engineering: Chip-Scale Integration

May 2027

Virginia Tech, Blacksburg, Virginia

SKILLS AND INTERESTS

Technical Skills	Technical Schematic Design, PCB Layout, SMD Soldering, CAD, FDM 3D Printing
Software	Altium Designer, MATLAB, Fusion360, AutoCAD, MS Office, Git/GitHub/GitLab, Quartus, I
Programming	C, C++, Java, JavaScript, HTML/CSS, Python

EXPERIENCE

VT BOLT

September 2024 – Present

Junior Electrical Team Lead

Blacksburg, Virginia

- Led hardware development and electrical systems for a multidisciplinary engineering team.
- Assisted in technical leadership for a 60-person organization, providing design direction and consultation.
- Managed electrical and mechanical system documentation.
- Planned, executed, and led testing for battery and controls subsystems.

Hume Research Center

September 2024 – May 2025

Night's Watch Simulation Design Team

Blacksburg, Virginia

- Developed MATLAB simulations for UAV/UAS flight path modeling and obstacle avoidance.
- Implemented Python-based drone simulation environment using VisPy for real-time visualization.
- Collaborated on AI-integrated counter-UAV research simulations.

PROJECTS

A2C (Anything To CAN)

- Designed PCB utilizing an STM32 to adaptively interpret sensor inputs and transmit data over a CAN bus.
- Supported SPI, I2C, UART, and analog sensor interfaces for universal sensor integration.

Battery Testbench

- Led a four-person team in system design and design integration.
- Designed STM32-based PCB for battery cell discharge, temperature, and current measurement.
- Integrated custom power management circuitry using the STM32 L9963E BMIC.

Internal Wiki

- Built custom React.js frontend for internal team knowledge cataloging.
- Developed MongoDB backend infrastructure.

ACHIEVEMENTS

2024 Technology Student Association Nationals – Structural Engineering and Design, 3rd Place

2023 Technology Student Association Nationals – Structural Engineering and Design, 1st Place

2022 Technology Student Association Nationals – Structural Engineering and Design, 3rd Place

PORTFOLIO WEBSITE

ABLOOM.CC